

**YEAR 10 SCIENCE
BIOLOGICAL SCIENCES
REVISION**

1. (a) Give a brief description of the DNA molecule.

• Double helix of sugar-phosphate.
• Bases join them.

- (b) Name the **4 bases** in a DNA molecule.

Guanine Cytosine Adenine Thymine

- (c) Name the **base pairs** found in DNA.

G - C A - T

- (d) Draw the adjoining strand of DNA for the following.

- G - T - T - A - C - A - G - T
- C - A - A - T - G - T - C - A

2. (a) Which process of cell division is responsible for the growth and repair of cells?

Mitosis

- (b) Which process is involved in producing the reproductive cells of an animal?

Meiosis

- (c) What is the meaning of **haploid** and **diploid number**?

Haploid - half normal chromosome number.

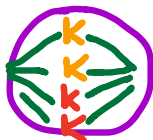
Diploid - normal number

3. An organism has 4 chromosomes in its normal body cells. Draw a series of simple diagrams showing the stages involved in **mitosis**. Name each stage and state what is happening in each.



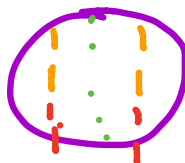
Prophase

Chromosomes
become visible
+
duplicate



Metaphase

Chromosomes
line up across
equator.



Anaphase

Chromosomes are
pulled apart.



Telophase
Cell divides into
two.

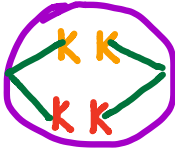
4. Meiosis occurs in reproductive organs in **two stages**. Draw a series of diagrams to show the stages involved. **Name each stage** and state **what is happening** in each.

STAGE I



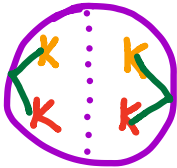
Prophase I

Chromosomes become visible + duplicate



Metaphase I

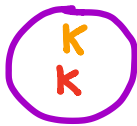
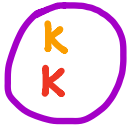
Line up in pairs!



Anaphase I

Pairs separate.

Now mitosis occurs!



Telophase I 2 cells with haploid number.

5. (a) What is the difference between the male and female sex chromosome pair?

XX - female XY - male

- (b) Who determines the sex of a child - the mother or father? Explain.

Father - Y chromosome determines maleness.

6. (a) Show, by punnett square, why half of any particular population would be expected to be female.

	♂		
♀	X	Y	
	X	XX	XY
	X	XX	XY

$\underbrace{XX \quad XX}_{50\% \text{ female}} \quad \underbrace{XY \quad XY}_{50\% \text{ male}}$

- (b) A family has had five girls born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl? Explain why.

50% $\frac{1}{2}$ of male sperm is X,
 $\frac{1}{2}$ is Y.

7. Briefly describe the significance of the work of Gregor Mendel to our understanding of genetics.

Grew pea plants. Noted that some characteristics dominate others.

8. Define the following terms.

allele one form of a gene. e.g. T, t

pure-bred alleles the same e.g. TT, tt

hybrid alleles different e.g. Tt

homozygous alleles same

heterozygous alleles different

genotype genes for a characteristic e.g. TT

phenotype how it looks e.g. Tall

sex-linked disorders carried on the X or Y chromosomes

dominant gene that masks another e.g. Tall - T

recessive gene that is hidden e.g. short - t

co-dominant neither gene dominates e.g. RW - red + white both show

incomplete dominance a new trait shows e.g. pink

haploid half the number of chromosomes.

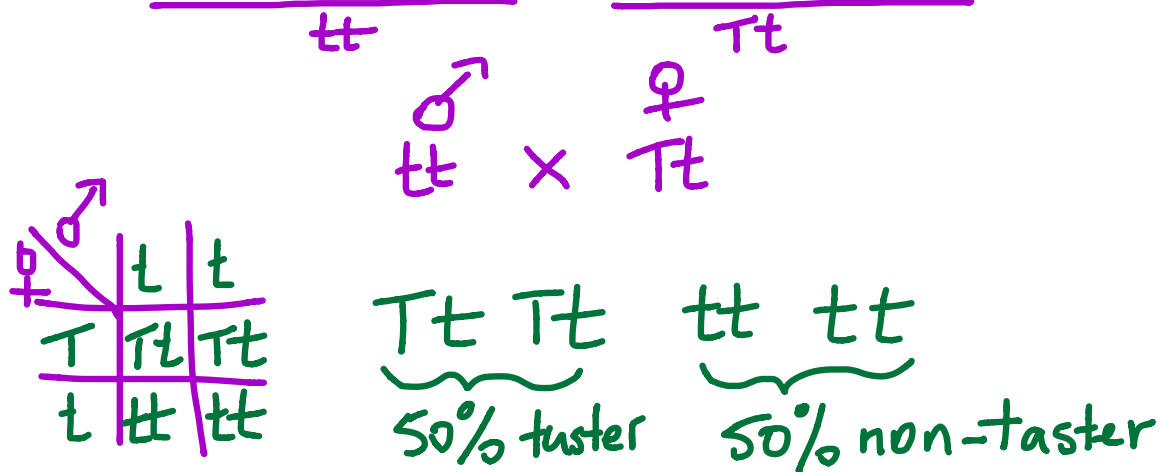
diploid. normal number of chromosomes.

9. Give one example of how **environment** can affect how a characteristic shows in an individual.

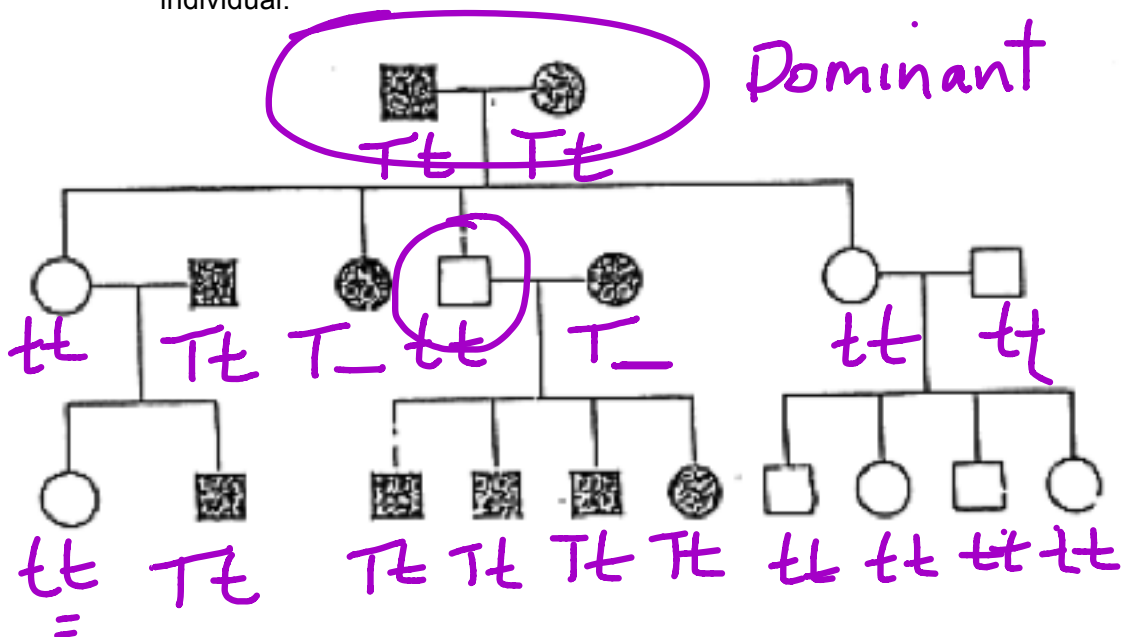
Person has blonde hair (genetic) – dyes it red (environmental)

10. In humans, the ability to taste a chemical phenylthiocarbamide (PTU) is due to the presence of the dominant gene T.

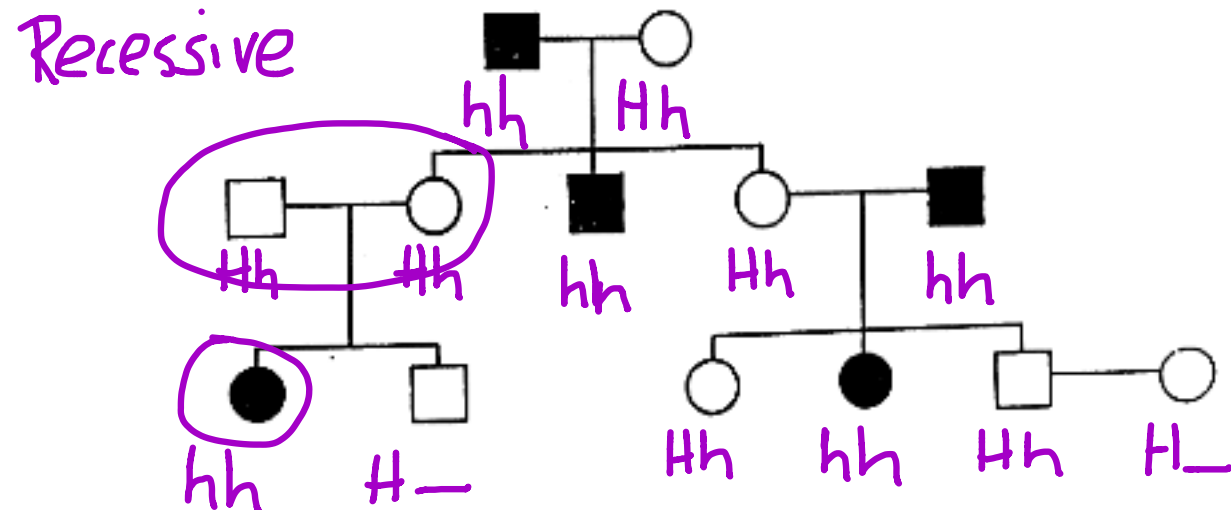
- (a) Determine the genotype and phenotypes of the offspring produced by the marriage of a homozygous non-taster male and a heterozygous taster female.



- (b) Study the pedigree chart below showing the occurrence of tasters in a family. A shaded square or circle represents a taster. Determine the genotypes of every individual.



11. The pedigree below shows the existence of straight hair in a family.



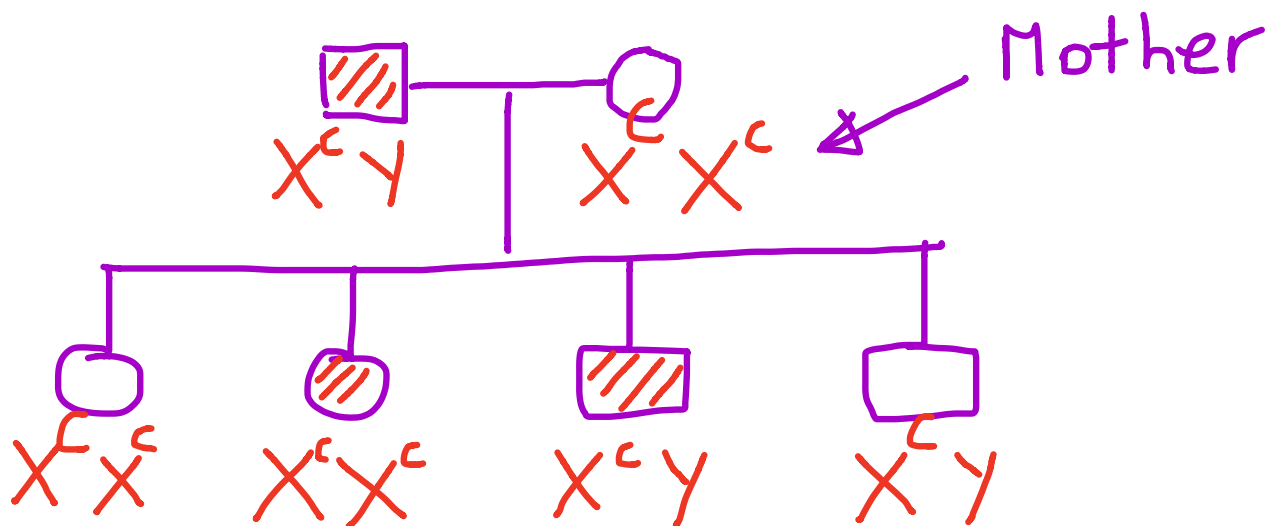
- (a) Is straight hair a dominant or recessive trait? Explain your answer.

Recessive. 2 parents don't have the trait but 1 child does.

- (b) Determine the genotype of all individuals.

12. Red-green colour blindness is a sex-linked, recessive disorder in humans. A colour-blind male marries a woman with normal colour vision. They have two girls, one of whom is colour blind, and two boys, one of whom is colour blind.

What is the most likely genotype of the mother? Set out an appropriate cross or pedigree to support your answer.



13. (a) Explain the difference between **co-dominance** and **incomplete dominance**. You may need to give an example of each to help your explanation.

Co-dominance - both traits show e.g. RW - red + white
 Incomplete - new trait shows e.g. RW - pink

- (b) In shorthorn cattle, red coat colour co-dominates with white coat colour, producing a roan colour in a heterozygous animal. Determine the F1 and F2 generation offspring for a red bull mated with a white cow. Set out your working.

♂ RR × ♀ WW

♂ \ ♀	R	R
W	RW	RW
W	RW	RW

F₁ RW RW RW RW
 100% roan.

RW × RW

♂ \ ♀	R	W
R	RR	RW
W	RW	WW

F₂ RR RW RW WW
 25% red 50% roan 25% white

- (c) In snapdragon flowers, red colour is incompletely dominant over white. Hybrid plants have pink flowers. Determine the genotypes and phenotypes for the offspring produced from a red flowered plant with a pink one.

♂ RR × ♀ RW

♂ \ ♀	R	R
R	RR	RR
W	RW	RW

RR RR RW RW
 50% red 50% pink

14. In guinea pigs, rough coat dominates smooth coat. Determine the genotypes and phenotypes for the mating of two rough coated hybrid animals.

Handwritten Punnett square and results for guinea pig coat color:

Parents: $Rr \times Rr$

$\begin{matrix} \text{♀} \\ \text{♀} \end{matrix}$	$\begin{matrix} \text{♂} \\ \text{♂} \end{matrix}$	R	r
R		RR	Rr
r		Rr	rr

Results: RR Rr Rr rr
 75% rough 25% smooth

15. In humans, suppose that the gene that controls right-handedness is dominant over left-handedness. A heterozygous right-handed male and a left-handed female have four children, three of who are left-handed.

- (a) If R represents the gene for right-handed, use a Punnett square to determine the expected ratio of children that should be right-handed.

Handwritten Punnett square and results for human handedness:

Parents: $Rr \times rr$

$\begin{matrix} \text{♀} \\ \text{♀} \end{matrix}$	$\begin{matrix} \text{♂} \\ \text{♂} \end{matrix}$	R	r
r		Rr	rr
r		Rr	rr

Results: Rr Rr rr rr
 50% right 50% left

- (b) Suggest a reason for difference between the result in part (a) and the actual proportion of children in this family who are right-handed.

Chance - there is a 50/50 chance of R or r in the gametes.

16. (a) Define the term **adaptation**.

Special feature that allows an organism to survive better

- (b) There are three types of adaptations - give an example for each.

(i) Behavioural Kangaroos sleep during the day.

(ii) Structural Bats have wings / large ears.

(iii) Physiological Koalas can eat eucalyptus leaves.

17. Explain how each of the following examples are adaptations for survival.

- (a) Desert marsupials excrete small quantities of very concentrated urine.

Conserve H_2O .

- (b) Polar bears have a thick fur coat.

Provides insulation

- (c) Frogs burrow into the ground during hot and dry weather.

Remain moist

18. Why is **genetic diversity** so important to the survival of a species?

Provides different phenotypes that help the species to adapt and survive.

19. What is the relationship between **mutation** and **genetic variation**?

Mutation changes genetic information $\Rightarrow$ variation occurs.

20. What is meant by the term **natural selection**?

Survival of the fittest - best adapted organisms survive

21. Myxomatosis virus was used to control rabbits in Australia but it is no longer effective. Explain why this is the case.

Some rabbits were resistant (low %).
These passed their genes on so a greater % are resistant.

22. Describe how **variation**, **selection pressures** and **competition** contribute to natural selection, and hence evolution.

Variation – produces traits that improve survival.

Selection pressures – living + non-living factors select which traits survive.

Competition – those with stronger traits win the competition for food, H_2O , light, space, etc.